Pandit Deendayal Energy University
School of Technology
Department of Computer Science and Engineering
Odd Semester 2025-2026
Course student handout file
INDEX

Name of the course: Introduction to Web Technology	**Course Code:**23CP306T
Program: B.Tech. **Branch: CSE**	**Semester:5th** **Academic Year: 2025-26**
Name of Course Coordinator: Dr. Debabrata Swain	
Subject Teachers: Dr. Punit Gupta, Dr. Nandini Modi, Prof. Komal Singh	

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Date:

Signature of Subject Teachers	**Signature of Department Coordinator (IQAC)**	**Signature of Head of the Department**

A. Departmental Vision & Mission

Vision

"To contribute to the society by imparting transformative education and producing globally competent professionals having multidisciplinary skills and core values to do futuristic research & innovations."

Mission

- To accord high quality education in the continually evolving domain of Computer Engineering by offering state-of-the-art undergraduate, postgraduate, doctoral programmes.
- To address the problems of societal importance by contributing through the talent we nurture and research we do:
- To collaborate with industry and academia around the world to strengthen the education and multidisciplinary research ecosystem.
- To develop human talent to its fullest extent so that intellectually competent and imaginatively exceptional leaders can emerge in a range of computer professions.

B. Program educational objectives (PEOs) of Department

The Program Educational Objectives of B.Tech. (Computer Engineering) program are:

1. To prepare graduates who will be successful professionals in industry, government, academia, research, entrepreneurial pursuit and consulting firms
2. To prepare graduates who will make technical contribution to the design, development and production of computing systems
3. To prepare graduates who will get engage in lifelong learning with leadership qualities, professional ethics and soft skills to fulfill their goals
4. To prepare graduates who will adapt state of the art development in the field of computer engineering

C. Program Outcomes (POs)

Undergraduate engineering program are designed to prepare graduates to attain the following program outcomes:

1. Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
2. Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
3. Design / development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with

appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

4. Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
6. The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
11. Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

D. Program Specific Outcomes (PSOs)

The graduates of CSE department will be able to:

1. Develop computer engineering solutions for specific needs in different domains applying the knowledge in the areas of programming, algorithms, hardware-interface, system software, computer graphics, web design, networking and advanced computing.
2. Analyze and test computer software designed for diverse needs.
3. Pursue higher education, entrepreneurial ventures and research.

E. Academic Calendar

PANDIT DEENDAYAL ENERGY UNIVERSITY

Academic Calendar: 2024-25

Even Semester: UG Sem.2/4/6/8 (FoET & FoLS) & PG Sem. 2/4 (FoET & FoLS)	
Particulars	**Date**
Semester Registration, Department Orientation & Commencement of classes.	6th Jan 2025
Republic Day Celebration	26th Jan 2025
Energy Cup – The Sports Fest	10th -14th Feb 2025
Internal Assesment-1 (Quiz, Test, Assignment etc.)** CP / MP Review – 1, Student Mentoring Week - 1	3rd -7th Feb. 2025
Attendance Review-I	24th -28th Feb. 2025
Mid Semester Examination / Dissertation Review 1	3rd March 2025 onwards
Last date of showing evaluated answer books and submission of mid semester exam marks	24th March 2025
Declaration of Mid Sem. Exam Result	28th Mar 2025
Attendance Review-II	7th -11th April 2025
Students Feedback	31st March - 4th April 2025
Declaration of detention list of students	21st April 2025
Internal Assesment-2 (Quiz, Test, Assignment etc.)** CP / MP Review – II Student Mentoring Week - 2	7th-11th April 2025
Flare – The Social & Cultural Fest	4-6th April 2025
Classes End & Farewell by Dept. to pass out batch	25th April 2025
Practical Examinations, submission of Term Work and Seminars	28th April to 3rd May 2025
Dissertation submission for UG and PG for FoLS	28th April 2025 onwards
End Semester Examination FoET & FoLS	7th May 2025 onwards
Dissertation Exam for PG program(s),CP / MP Final Review	19th - 23rd May 2025
Last date of showing evaluated answer books and submission of end semester exam marks	30th May 2025
Internship for FoET students (Industrial/ Rural/ CSSI)	May-July 2025
Progress Review for Ph.D.	May- June 2025
Declaration of End Sem. Exam Result	6th June 2025
Summer Vacation (Faculty will resume office on **9th July 2025**)	2nd June - 8th Jul 2025
Faculty Development Programme	14th -18th July 2025
Odd Semester: FoET & FoLS	
Odd Semester Registration & class commencement	21st Jul 2025

F. Class Time Table and Faculty Time Table with office hours

Class Time Table

Div 5

Faculty Time Table : Dr. Debabrata Swain

Debabrata Swain
Computer Science & Engineering

Even Semester 2025, w.e.f : 6th January 2025[Printed : 19-12-2024]

Day	08:00-09:00	09:00-09:55	10:00-10:55	11:10-12:05	12:10-13:05	13:10-14:05	14:10-15:05	15:10-16:05	16:15-17:10	17:15-18:10
Monday		G3G4 (20CP208T) F-402, CP(4) - L		G4 (20CP208P) E216, CP(4) - P						
Tuesday				G9 (23CP308P) F-104, CP(6) - P			G7 (20CP210P) E213, CP(4) - P			
Wednesday			G3G4 (20CP208T) F-402, CP(4) - L	G7 (20CP210P) E215, CP(4) - P						
Thursday				G3 (20CP208P) F-203, CP(4) - P						
Friday				G3G4 (20CP208T) F-302, CP(4) - L	G3G10 (23CP308T) F-302, CP(6) - L					

Location Abbr.	Location Name	Subject Abbr.	Subject Name
E213	E, Web Tech. Lab (CP2)	20CP208P	Database Management System - Lab
E215	E, Data Struct. & Algo Lab	20CP208T	Database Management System
E216	E, DBMS LAB(CNSLAB)	20CP210P	Design Pattern - Lab
F-104	F, Data Analytics Lab	23CP308P	Advanced Web Technology - Lab.
F-203	F, Security & Comp. Lab	23CP308T	Advanced Web Technology
F-302	F, Lecture Hall		
F-402	F, Lecture Hall		

Office Hours: Wednesday 4:00 to 5:00 pm

G. Course Outcomes (COs), Course Syllabus, Pre requisites for the course

Pandit Deendayal Energy University School of Technology

Subject Name/Lab Name					Advanced Web Technology Theory					
Teaching Scheme					Examination Scheme					
L	T	P	C	Hrs/Week	Theory			Practical		Total Marks
					MS	ES	IA	LW	LE/Viva	
0	1	0	1	1	25	50	25	-	-	100

COURSE OBJECTIVES

- To introduce the advanced web programming technologies.
- Understand Client Side Web Development

UNIT 1 Web Development with Node and Express The JavaScript Revolution, Introducing Express, A Brief History of Express, The Node Ecosystem, What is Express.js, How Express.js Works, MVC Structure and Modules	**5 Hrs.**
UNIT 2 React's Past and Future, How React Works, React State Management	**4 Hrs.**
UNIT 3 Django Intro, MVT Architecture, Views, URL, Templates, Models, Database Connectivity (PostgreSQL)	**5 Hrs.**
UNIT 4 Flask Introduction, Flask- Application, Routing, Variable Rules, URL Building, HTTP methods,Templates, SQLAlchemy	**3 Hrs.**
	Max. 17 Hrs.

COURSE OUTCOMES

On completion of the course, student will be able to
CO1- Explain the concepts of Web Application
CO2- Identify and explain various components of Web Application stack
CO3- Illuminate technologies and tools involved in Web Application Development
CO4- Understand and Identify the use of Web Services in Web Programming
CO5- Learn Javascript based runtime environments: NodeJS, ReactJS.
CO6- Understand Python frameworks for web development: Django and Flask.

TEXT/REFERENCE BOOKS

- Django for Beginners: Build Websites with Python and Django by William S. Vincent
- Flask Web Development: Developing Web Applications with Python, Second Edition (Greyscale Indian Edition) by Miguel Grinberg
- React and React Native: A complete hands-on guide to modern web and mobile development with React.js, 3rd Edition by Adam Boduch and Roy Derks
- Beginning MERN Stack: Build and Deploy a Full Stack MongoDB, Express, React, Node.js App by Greg Lim

END SEMESTER EXAMINATION QUESTION PAPER PATTERN

Max. Marks: 100 **Exam Duration: 3 Hrs**

Part A: 10 Questions of 2 marks each-No choice	20 Marks
Part B: 2 Questions from each unit with internal choice, each carrying 20 marks	80 Marks

H. Lesson Plan

Lecture No.	Topic to be covered	Teaching Aid to be used	Remarks (Text book/Unit No etc.)
1	The JavaScript Revolution	BW+PPT	Unit 1
2, 3, 4	Introduction to node, The Node Ecosystem		
5	Introducing Express		Unit 2
6	What is Express.js		
7	MVC Structure and Modules		Unit 3
8	How React Works,		
9	React hooks		
10	React State Management		
11	Django		
12	Intro, MVT Architecture, Views, URL, Templates, Models,		Unit 4
13	Database Connectivity (PostgreSQL)		

Legends : BW (Board Work), PPT (PowerPoint Slides)

I. Program Articulation Matrix and Course Articulation Matrix

Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1															
CO2															
CO3															
CO4															
CO5															
CO6															

Program Articulation Matrix

PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PS01	PSO2	PSO3

Correlation levels 1, 2 or 3 as defined below:

1: Slight (Low)

2: Moderate (Medium)

3: Substantial (High)

J. Evaluation Schem and Rubrics

CO Assessment Tools (Direct Assessment):
Various assessment tools used to evaluate CO's (Rubrics) and the frequency with which the assessment processes are carried out are listed below.

Assessment Method	Assessment Tool	Description	Marks	Mapping with CO	Contribution to CO's
Direct	Mid-sem	MCQ/Analytical/ Output-based/ questions on syllabus covered from Unit I, Unit II	25	CO1/CO2/CO3	It fractionally contributes to 100% weightage of Direct Assessment to CO attainment.
Direct	MCQ/Class Assignment	MCQ/ Analytical/ Output-based/ Theoretical questions on syllabus covered	25	CO1/CO2/CO3 CO4/CO5/CO6	It contributes to 100% weightage of Direct Assessment to CO attainment.
Direct	End-Sem Examination	**Topics to be covered:** Unit I, II, III, IV	100	CO1,CO2, CO3,CO4, CO5, CO6	It contributes to 50% weightage of Direct Assessment to CO attainment. (100/2)
	Total 100 Marks				

K. Tutorials, Assignments, Case Studies, Quiz, Presentations etc.

Attached herewith.

L. Copy of Mid and End Semester Examination Question Papers (Old and Current), solution of current examination with stage-wise marking scheme – Not applicable as this course is introduced in form of theory also after long time…

M. Course covered beyond syllabus

N. Attendance Record (Up to Mid Semester Examination and Up to End semester Examination)

O. Details for Remedial Classes (list and identification of slow learners, actions taken)

P. Result of students (marks of mid, end and internal assessment components) – Attached herewith.

Q. Laboratory Manuals – If Applicable (NA)

R. List of International / National Journals Related to the Course – If Applicable

1. International Journal of Web Engineering and Technology (https://www.inderscience.com/jhome.php?jcode=ijwet)
2. International Journal of Information Technology and Web Engineering (IJITWE) (https://www.igi-global.com/journal/international-journal-information-technology-web/1093)
3. International Journal of Web Information Systems (https://www.emerald.com/insight/publication/issn/1744-0084)

S. List of Well-known Conferences Related to the Course – If Applicable

1. Web Summit
2. TechCrunch Disrupt

T. Sample Answer Scripts of Mid Sem., End Sem. Exam and Assignments of Good, Better, and Best Performing Students (At Least Five Copies of Each Assessment Tool) (NA)

U. Class Notes (Lecture PPT & Lab Manual etc.) in Soft/Hard Copy
Attached herewith.

V. Direct Attainment of COs and POs and Interpretation (Result Analysis)

Attached herewith.

W. Indirect Attainment of POs through Course Exit Survey (Just Before End Sem. Exam)

Attached herewith.

X. Final Attainment of COs and POs and Interpretation (Result Analysis), Actions to be Taken if COs and POs are Not Achieved

N/A

School of Technology, Pandit Deendayal Energy University, Gandhinagar

Department of Computer Science & Engineering

Expert Lecture Report

Name of the Expert	
Affiliation of the Expert	
Contact Details of the Expert (Address, E-mail, Phone, etc.)	
Title of the Lecture	
Date of Lecture	
Duration of Lecture	
Audience (B.Tech., M.Tech. with Semester)	

Venue	
No. of Attendees	
Feedback on the Lecture (if any)	
Photograph of the Interaction / Meeting /Lecture	
Name of the Faculty Coordinator (who arranged the lecture)	
Signature of the Faculty Coordinator	